

## Performance Testing

|                      |                                       |
|----------------------|---------------------------------------|
| <b>Date</b>          | 01 JULY 2026                          |
| <b>Team ID</b>       | XXXXXX                                |
| <b>Project Name</b>  | A Comprehensive Measure Of Well-Being |
| <b>Maximum Marks</b> | 5 Marks                               |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field                      | Details                                               |
|----------------------------|-------------------------------------------------------|
| <b>Testing Tool Used</b>   | Browser Testing (Google Chrome), Flask Local Server   |
| <b>Type of Testing</b>     | Response Time Testing, Functional Performance Testing |
| <b>Target Module / API</b> | Well-Being Prediction Module                          |
| <b>Test Environment</b>    | Hosted Server (Render) & Local Machine                |
| <b>Test Date</b>           | 01 JULY 2026                                          |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description       | No. of Virtual Users | Duration (sec) | Expected Outcome                                 |
|------|-----------------------------------|----------------------|----------------|--------------------------------------------------|
| 1    | Open Home Page / Get Started Page | 1                    | 10 sec         | Home Page / Get Started Page loads successfully. |
| 2    | Enter Assessment Values           | 1                    | 20 sec         | Input accepted without errors.                   |
| 3    | Predict Well-Being Score          | 1                    | 15 sec         | Prediction displayed successfully.               |
| 4    | Generate Recommendations          | 1                    | 15 sec         | Personalized recommendations displayed.          |

### Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks                 |
|------|----------------------|--------------|--------------|----------------------|-------------------------|
| 1    | Response Time (Avg)  | < 2 seconds  | 1.3 seconds  | Pass                 | Good Response time      |
| 2    | Response Time (Max)  | < 5 seconds  | 2.4 seconds  | Pass                 | Within acceptable limit |
| 3    | Throughput (Req/sec) | 5 Req/sec    | 6 Req/sec    | Pass                 | Stable                  |
| 4    | Error Rate           | < 1%         | 0%           | Pass                 | No Error observed       |
| 5    | CPU Utilization      | < 80%        | 35%          | Pass                 | Normal usage            |
| 6    | Memory Utilization   | < 80%        | 42%          | Pass                 | Stable                  |

### Step 4: Observations & Analysis

#### Key Findings:

The application responded quickly to user inputs and successfully generated well-being predictions and recommendations. No significant delays were observed during testing.

#### Bottlenecks Identified:

No major performance bottlenecks were identified during testing. Initial loading on Render may take a few seconds because the free instance wakes up after inactivity.

#### Optimization Steps Taken:

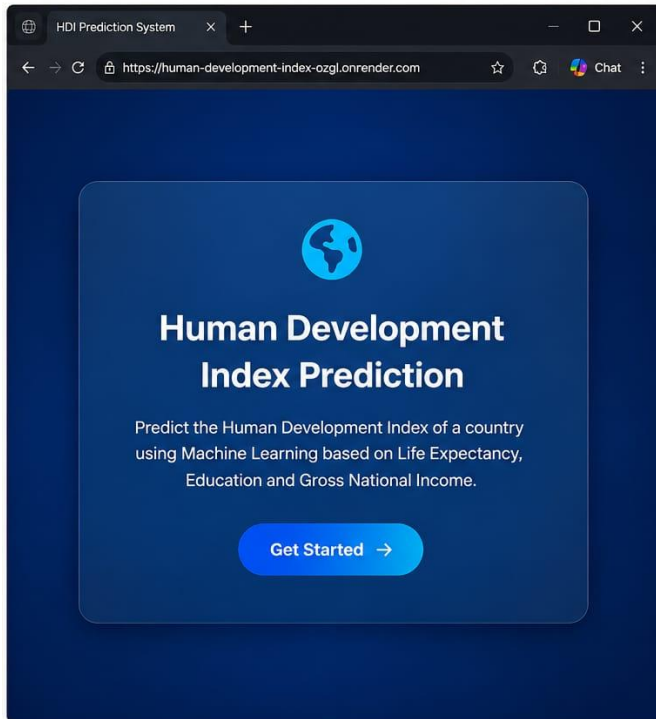
- Optimized the Flask application for efficient request handling and improved performance.
- Used a trained Pickle model to enable faster and efficient predictions.
- Reduced unnecessary processing during prediction to improve response time.
- Deployed the application on Render for reliable online accessibility and testing.

### Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

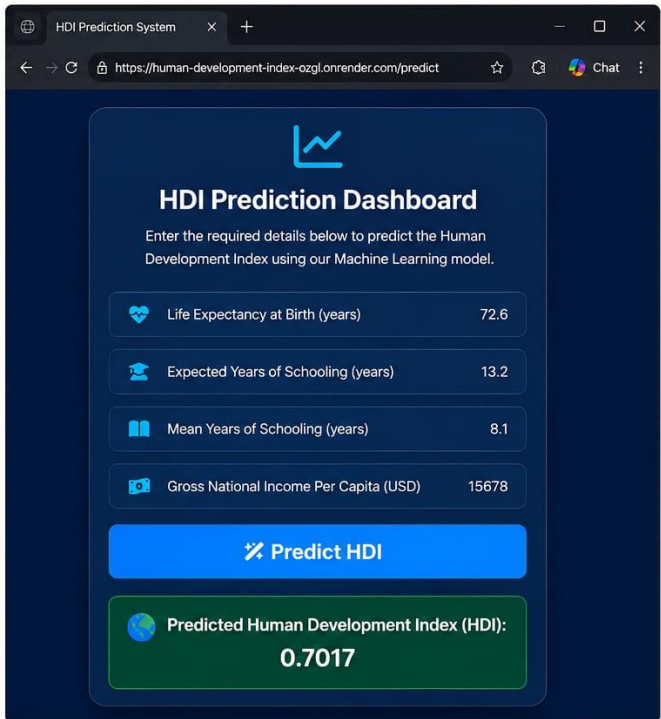
## Step 5: Screenshots / Evidence

Screenshot 1: Home Page (Get Started Screen)



Home / Get Started Page

Screenshot 2: HDI Prediction Dashboard (Result Screen)



HDI Prediction Result